Beyond the Monolith: Open Middleware for Heterogeneous AI Infrastructure at Hyperscale

Gaurav Agarwal (mod)

Senior Distinguished Engineer, Marvell

Panelists:

Chris Gottbrath - Gottbrath Technologies

Mayank Saxena - Samsung

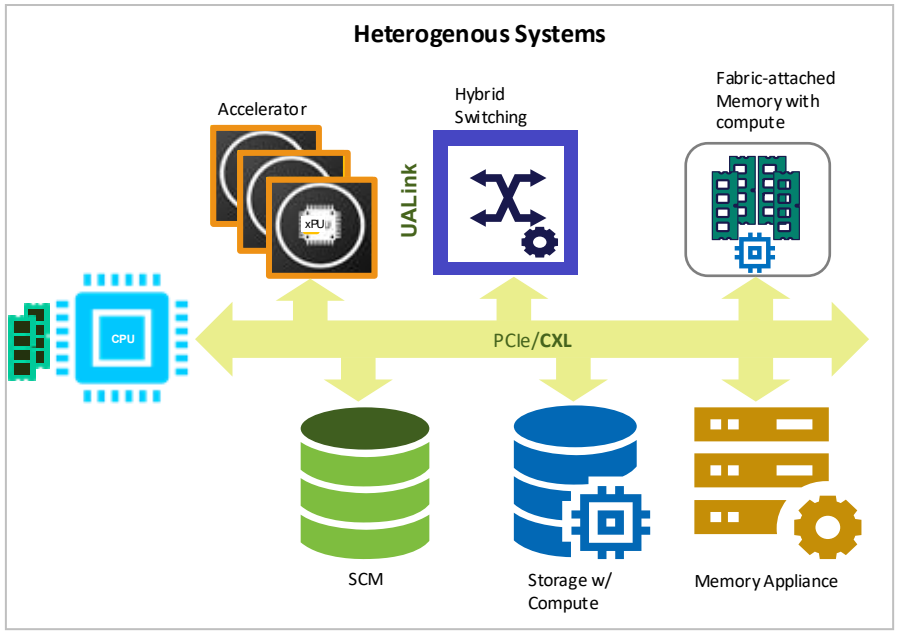
Harry Kim - XCENA

Youngpyo Joo - SK hynix

OCP FUTURE TECHNOLOGIES SYMPOSIUM

October 14, 2026
San Jose, California

Context: software for heterogenous computing



Software Challenges

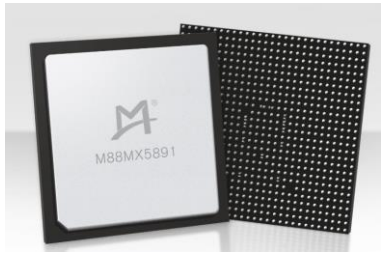
- Heterogenous Programming Model
- Disaggregated Computing
  - Heterogeneity-aware compute partitioning
  - Deep-memory hierarchies
  - Topology variability
- Control Plane (Operational Complexity)
  - Discovery
  - Configuration
  - Resource provisioning

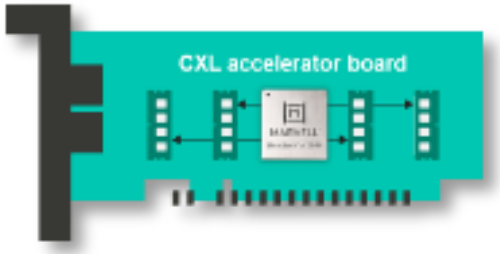
Direction

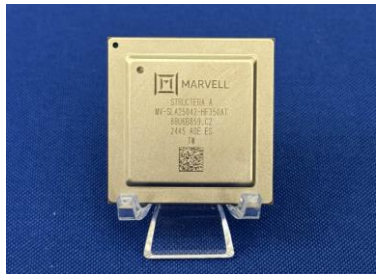
- Software for bridging AI frameworks and Computational Memory and Storage Devices (CxDs)
  - A Stable, Performant, and Portable API
  - Ensure multi-vendor interoperability
- Portable Performance on AI Data Sets
  - Function and Data Placement optimizations
  - Zero-copy Arrow open formats for vector data & ops
  - Mix of Graph and Tensor Data in large VA objects
- Open Format & Memory Objects
  - Uniform interfaces to Comp. Mem., Comp. Storage, and DPUs
  - Uniform specification of offloads for Data at rest or Data in flight

Heterogenous compute continues to evolve

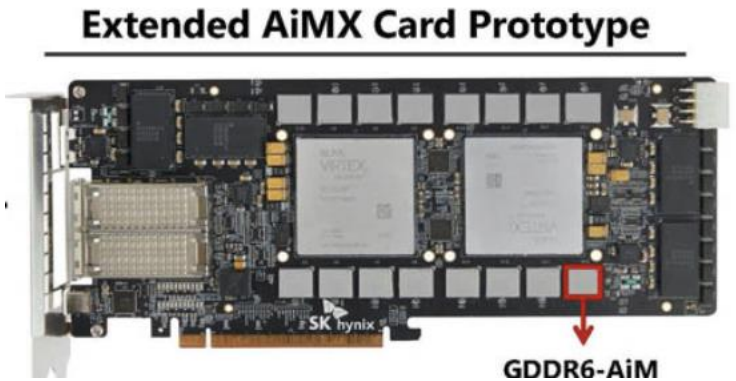


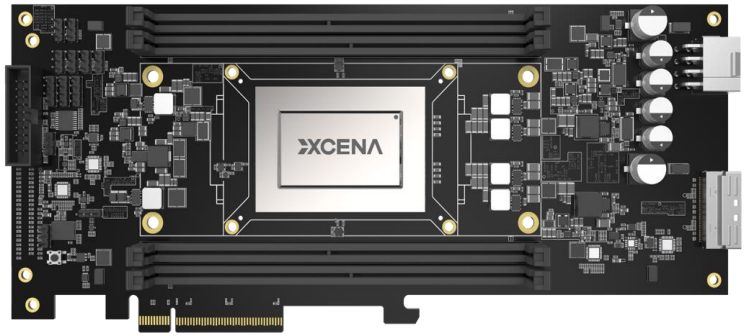


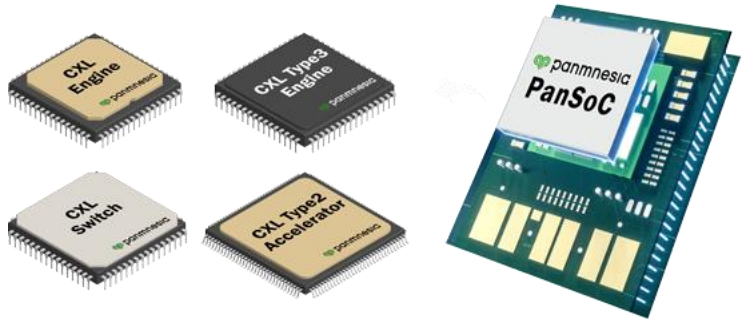












From prototypes to datacenter ready hardware is already available; and growing!

Portable software for heterogenous AI Infra

Open frameworks that use graph lowering compilers and multi-level intermediate representations

AI Models | **Framework + runtime** | **Programming model** | **Disaggregated infrastructure**



- Hardware agnostic
- Portable performance
- HW-independent innovation

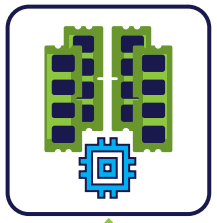
Many considerations
- Optimal placement
- Access patterns
- Operator fusion
- …

- Domain optimizations
- Target optimizations
- Auto-sharding
- Orchestration

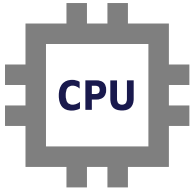
Pluggable backends

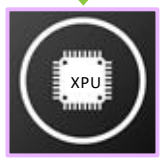
Device-specific optimizations





Memory with compute





- Latency/Power/Cost-optimized Programmable Compute
- Specialized fixed-function compute
- SW-independent innovation

Needs cross-industry collaboration



Chris Gottbrath

Principal - Gottbrath Technologies

Active in the PyTorch community
Ex-Meta, Ex-NVIDIA

OCP FUTURE TECHNOLOGIES SYMPOSIUM

October 14, 2026
San Jose, California

AI Ecosystem Challenges

- Forces driving pressure for more memory
  - Context Window
  - Model Size / Expert Parallelism
  - Agentic Tool Use
- More specific motivating constraints for different parts of the life cycle
  - Inference: Latency/time to first token, RAG & tool use, user context, serving cost at scale
  - Pre-training: Throughput, fault resilience, highly distributed architectures, efficiency at cluster scale
  - Post-training: Reinforcement learning, tool use, scheduling heterogenous jobs
- Heterogeneity and disaggregation are already a part of the architecture
  - In an already complex system - there is a cost to adding another tier to schedule and optimize around
  - Expect questions from maintainers who want to keep the core code paths stable and clean
    - Hardware availability, active maintainers, CI/CD, testing, and performance regression testing

Integration points within PyTorch

- PyTorch has a rich set of options for prototyping and integration
- Python/C++ based extensions
  - Subclass the memory allocator
  - Provide custom operators that can be called directly in prototype code
    - Triton and TLX for kernel authoring
- PrivateUse1 dispatch key
  - Intended for out-of-core-repository accelerators but might be useful in this context as well
  - Allows allocation and custom operator overrides with fallback to native
  - Pytorch Compiler integration
    - High Level: Dynamo
    - Low Level: Inductor, Triton
  - OpenReg is a (new) reference implementation
  - PrivateUse1 is leveraged by IBM Spyre, Huawei Ascend, and Cambricon back ends.
  - *Potential Challenge* we might need a new framework concept to compose non-uniform memory with compute
- Other integrations to look at: AMD, Intel XPU, Apple MPS, and Google TPU

Path to hardware-software co-design

- Think in terms of reusable patterns
  - Identify key computational patterns with clear applicability
  - repeatable architectural patterns
- Lead with proof points
  - Show what kind of performance is possible, code can be hacky
  - Measure addressable gaps and show gains that could be delivered with more optimization
- Create space for collaboration
  - Hardware availability
  - Forums where software, hardware, and potential users can all be part of the discussion

Mayank Saxena

Chief Technologist, Memory Solutions Lab - Samsung



OCP FUTURE TECHNOLOGIES SYMPOSIUM

October 14, 2026
San Jose, California

“Heterogeneous HW” Co-Design

Open Base Interfaces | Specialized App Contracts | Improve Peer Value

Interoperability without slowing innovation

Standardize smaller, integrate sooner, velocity!

Stable APIs; HW fast paths evolve independently.

Example: same All-Reduce API, competing transports

Workload Centric Slice-ware

Traditional middlewares need to earn their latency & complexity

Expose HW capabilities through thin, workload-specific paths.

Standardize essential contracts, not execution path.

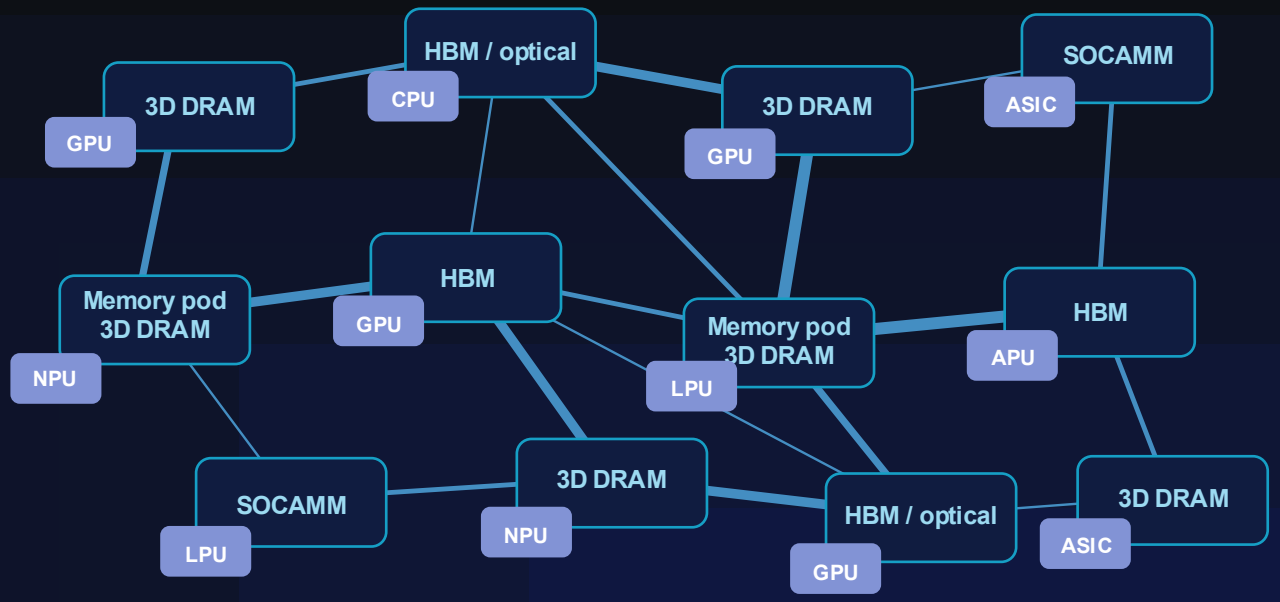
Industry parallel: Global Logistics

One container across ships, trains and trucks.

Standardize the handoff, not the vehicle.

Multiply peer value - Ships provide high-capacity; trucks provide flexible delivery.

Heterogeneous Memory-Compute Fabric



Link widths: illustrative capability / bandwidth differences; not to scale

Open Value Prop : One slice • All vertical stakeholders • Multiple implementations • Greater system multiplier

Example: Cognos, composing heterogenous AI memory fabric (coming soon: Open Source Oct ‘26)



Harry Kim

Chief Product Officer, XCENA

OCP FUTURE TECHNOLOGIES SYMPOSIUM

October 14, 2026
San Jose, California

Data Centric Computing – Data First Approach

Data Placement

Right data → right performance/cost tier is critical

- Hard even with just two tiers
- HBM · DRAM · CXL · SSD · network
- Mixed media and interconnects
- Answer differs by user and application

Benchmark Driven

Evidence before interfaces

- No device wins every workload
- Representative workloads → right placement decisions
- Before (or alongside) a unified API
- Beyond any single device vendor: needs collaboration

Data View / Memory Model

Different devices, different views

- Accelerator vs. CXL,
  PIM vs. in-storage compute,
  all different memory models
- Need to understand what differs, and why
- Basis for a real common platform

Otherwise, **unified middleware becomes a junk drawer**, everything in one place, nothing fits together.



Youngpyo Joo

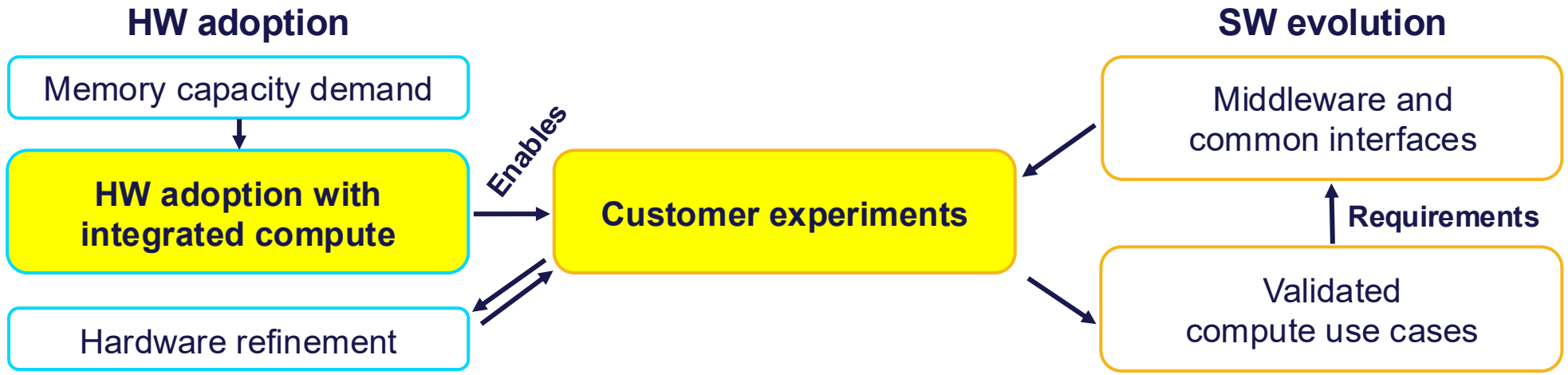
Vice President of System Architecture, Memory Systems Research - SK hynix

OCP FUTURE TECHNOLOGIES SYMPOSIUM

October 14, 2026
San Jose, California

Computational Memory: Adoption and Middleware Direction

:: Breaking the hardware adoption dilemma



:: Key challenges for middleware to address

Data placement across devices

Data required for computation may reside on a different device, challenging the conventional practice of moving data to the processor.

Hardware-specific data layout

Optimal data layouts vary across hardware, but software developers should not have to manage these differences.

OCP FUTURE TECHNOLOGIES SYMPOSIUM

October 14, 2026
San Jose, California

